

**A
REPORT
ON
FAKE NEWS DETECTING SYSTEM
By**

Name of the Students	Registration No:
P.Yashaswini	AP24110010959
B.Varshitha	AP24110010957
E.Vindhya Sri	AP24110010935
B.Sai Hasitha	AP24110010908

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CERTIFICATE

This is to certify that the project entitled FAKE NEWS DETECTION by P.Yashaswini(AP2424110010959), B.Varshitha(AP24110010957), B.Sai Hasitha(AP24110010908) and E.Vindhya sri(AP24110010935) as a required academic component of the course CSE 205: Hands-on with Python during Semester 4 of Academic Year 2025-26 in the Department of Computer Science and Engineering at SRM University, AP. This work was executed under the guidance of the undersigned and is deemed to have successfully met all course requirements as of April 2026.

Dr.sravan SS

Assistant Professor

Department of computer science and engineering

SRM university

ABSTRACT

The rapid growth of digital media platforms has significantly increased the spread of false and misleading information, commonly known as fake news. This creates serious challenges in identifying the authenticity of news content and can lead to misinformation among the public. To address this issue, this project presents a Fake News Detection system using techniques from Machine Learning and Natural Language Processing. The proposed system aims to automatically classify news articles as real or fake based on their textual content. A dataset containing labeled news articles is used to train the model. The text data undergoes preprocessing steps such as cleaning, normalization, and removal of unnecessary elements to improve accuracy. The processed text is then transformed into numerical form using feature extraction techniques, enabling the model to interpret the data effectively. A machine learning model is trained on this data to learn patterns and differences between real and fake news. Once trained, the system can predict the authenticity of new, unseen news articles provided by the user. The model demonstrates effective performance in distinguishing between genuine and fake information. This project highlights the importance of automated systems in combating misinformation and provides a foundation for further improvements using advanced models and larger datasets. The system can be extended for real-world applications such as social media monitoring and news verification platforms.

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1.Introduction

In the modern digital era, the rapid growth of the internet and social media platforms such as WhatsApp, Facebook, and Twitter has made the sharing of information faster and easier than ever before. However, along with the benefits of fast communication, there has been a significant rise in the spread of fake news. Fake news refers to false or misleading information presented as genuine news, which can create confusion, influence public opinion, and sometimes lead to serious social issues. Identifying whether a piece of news is real or fake has become a challenging task for individuals, as it requires time, effort, and proper verification.

Manual verification is not always practical, especially when large amounts of information are being generated and shared continuously. To address this problem, this project proposes a Fake News Detection System using Machine Learning. The system is designed to automatically analyze news content and classify it as either real or fake. By using historical datasets of real and fake news, the machine learning model learns patterns and features that distinguish between authentic and false information.

The main objective of this project is to develop an efficient and reliable system that can assist users in identifying fake news quickly and accurately. This system can help reduce the spread of misinformation and promote awareness among users.

2.Main Text

The Fake News Detection System is used to identify whether a news article is real or fake using machine learning. Nowadays, many people get news from the internet and social media, but not all the information is true. This system helps users to check the authenticity of news quickly.

The system uses a dataset that contains both real and fake news. First, the text is cleaned by removing unwanted symbols and converting it into lowercase. Then, the text is converted into numbers because computers can understand only numerical data.

After that, a machine learning model is trained using this data. The model learns the patterns of real and fake news. When a user enters new news text, the system processes it and gives the output as real or fake. The Fake News Detection System is used to identify whether a news article is real or fake using machine learning. Nowadays, many people get news from the internet and social media, but not all the information is true. This system helps users to check the authenticity of news quickly.

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This system is developed using Python and libraries like Pandas, NumPy, and Scikit-learn. It provides a simple and effective way to reduce the spread of false information.

3. System Methodology

The methodology of the Fake News Detection System explains the step-by-step process used to identify whether a news article is real or fake. The system follows a structured approach that includes data collection, preprocessing, feature extraction, model building, and prediction.

First, data is collected in the form of a dataset containing both real and fake news articles. This dataset is used to train the machine learning model. Each data entry includes the news text and a label indicating whether it is real or fake.

Next, the collected data undergoes preprocessing. In this step, the text is cleaned by removing punctuation, special characters, and unnecessary words. The text is also converted into lowercase to maintain consistency. This helps in improving the quality of the data.

After preprocessing, feature extraction is performed. Since computers cannot understand text directly, the cleaned text is converted into numerical form using techniques such as Count Vectorization or TF-IDF. This step transforms the text into a format suitable for machine learning algorithms.

Finally, the system is used for prediction. When a user enters a new news article, the same preprocessing and feature extraction steps are applied. The trained model then analyzes the input and provides the output as either real or fake.

Thus, the system follows a systematic approach to detect fake news efficiently and accurately.

3.1 Data collection

Data collection is the first and most important step in developing the Fake News Detection System. In this stage, a dataset containing a large number of news articles is gathered from reliable sources such as online news platforms, public datasets, and research repositories. The dataset consists of both real news and fake news examples, where each news article is clearly labeled as either “real” or “fake.”

The collected dataset plays a key role in training the machine learning model, as it provides the necessary input data for learning patterns and differences between genuine and misleading news.

A well-balanced dataset with a sufficient number of both real and fake news samples is essential to improve the accuracy and performance of the model.

Each record in the dataset typically includes two main components: the news content (text) and its corresponding label.

This structured format helps the system understand the relationship between the input (news text) and the output (classification result). Proper data collection ensures that the model is trained effectively and can perform well in real-world scenarios.

3.2 Data Preprocessing

Data preprocessing is the second and very important stage in the Fake News Detection System. In this step, the collected raw text data is cleaned and prepared for further processing. Real-world news data often contains noise such as punctuation marks, special characters, numbers, extra spaces, and inconsistent formatting, which can reduce the performance of the model if not handled properly.

During preprocessing, all text is first converted into lowercase to maintain uniformity and avoid duplication of words due to case differences. Then, unwanted symbols, punctuation, and special characters are removed. In some cases, common words (stop words) such as “is”, “the”, “and”, etc., are also removed because they do not contribute much meaning to the text.

This step helps in simplifying the dataset and improving the quality of the input data. By cleaning and standardizing the text, the system ensures that only meaningful information is passed to the next stage, which improves the accuracy and efficiency of the machine learning model.

3.3 Model Building

Model building is an important stage in the Fake News Detection System where a machine learning algorithm is selected and trained to classify news as real or fake. In this step, the preprocessed and feature-extracted data is used to create a predictive model that can learn patterns from the dataset.

For this system, commonly used algorithms such as Logistic Regression, Naive Bayes, or Support Vector Machine (SVM) are applied. These algorithms are well-suited for text classification tasks because they can effectively handle numerical representations of text data. The model is designed to understand the relationship between input features and output labels (real or fake).

During model building, the algorithm learns from the training data by identifying patterns and differences between real and fake news articles. The main objective is to create a model that can generalize well and make accurate predictions on new, unseen data. A well-built model improves the overall performance and reliability of the system.

3.4 Training and Testing

Training and testing is the final and very important stage of the Fake News Detection System. In this step, the prepared dataset is divided into two parts: training data and testing data. The training data is used to teach the machine learning model, while the testing data is used to evaluate its performance.

During the **training phase**, the model learns patterns and relationships between the input features and the output labels (real or fake news). It adjusts its internal parameters based on the training data to improve its accuracy. This helps the model understand how to distinguish between real and fake news effectively.

In the **testing phase**, the trained model is given new and unseen data to check how well it performs. The model predicts whether the news is real or fake, and these predictions are compared with the actual labels. Performance is measured using metrics such as accuracy, precision, and recall. These metrics help in evaluating how reliable and effective the model is.

This stage ensures that the model is not only trained properly but also capable of making accurate predictions on real-world data.

```
from flask import Flask, render_template, request
import pandas as pd
import re
import string
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

app = Flask(__name__)

def clean_text(text):
    text = str(text).lower()
    text = re.sub(r'["\'`]', '', text)
    text = re.sub(f"[{string.punctuation}]", "", text)
    text = re.sub(r'\s+', ' ', text)
    return text.strip()

# Load data
fake = pd.read_csv("Fake.csv")
true = pd.read_csv("True.csv")

fake["label"] = 0
true["label"] = 1

data = pd.concat([fake, true])

# Lookups
title_lookup = {}
text_lookup = {}

for _, row in data.iterrows():
    title_lookup[clean_text(row["title"])] = row["label"]
    text_lookup[clean_text(row["text"])] = row["label"]
```

```
# Prepare ML
data["content"] = data["title"] + " " + data["text"]
data["content"] = data["content"].apply(clean_text)

X = data["content"]
y = data["label"]

vectorizer = TfidfVectorizer(stop_words="english", max_df=0.8, min_df=2, ngram_range=(1,2))
X = vectorizer.fit_transform(X)

model = LogisticRegression(max_iter=1000)
model.fit(X, y)

# Route
@app.route("/", methods=["GET", "POST"])
def home():
    result = ""

    if request.method == "POST":
        news = request.form["news"]

        if len(news.split()) < 5:
            result = "⚠ Please enter more detailed news"
        else:
            cleaned = clean_text(news)

            if cleaned in title_lookup:
                label = title_lookup[cleaned]
                result = "🟢 Real News (Exact Match)" if label == 1 else "🔴 Fake News (Exact Match)"

            elif cleaned in text_lookup:
                label = text_lookup[cleaned]
                result = "🟢 Real News (Exact Match)" if label == 1 else "🔴 Fake News (Exact Match)"

    return result
```

```
        else:
            vector = vectorizer.transform([cleaned])
            prediction = model.predict(vector)[0]
            confidence = model.predict_proba(vector)[0]

            if max(confidence) < 0.65:
                result = "⚠ Not sure"
            elif prediction == 1:
                result = f"🟢 Real News ({confidence[1]*100:.2f}%)"
            else:
                result = f"🔴 Fake News ({confidence[0]*100:.2f}%)"

    return render_template("index.html", result=result)
if __name__ == "__main__":
    app.run(debug=True)
```

4. System Architecture

The Fake News Detection System is designed using a structured architecture that defines how data flows through different components of the system to produce the final output. The architecture mainly consists of five stages: input layer, preprocessing module, feature extraction module, machine learning model, and output layer.

The process begins with the **input layer**, where the user provides a news article as input. This input can be any text-based news content that needs to be verified as real or fake. The input data is then passed to the next stage for processing.

In the **preprocessing module**, the raw text is cleaned and standardized. This includes converting the text into lowercase, removing punctuation marks, special characters, numbers, and unnecessary words. This step ensures that the data is clean and consistent before analysis.

After preprocessing, the data moves to the **feature extraction module**. Since machine learning models cannot directly understand text, the cleaned text is converted into numerical format using techniques like Count Vectorization or TF-IDF. This transformation helps in representing the text in a structured form suitable for model processing.

Finally, the system produces an output in the **output layer**, where the model predicts whether the given news article is real or fake. The result is displayed to the user in a simple format, helping them quickly verify the authenticity of the news.

Thus, the system architecture ensures a smooth flow of data from input to output, enabling efficient and accurate fake news detection.

5.Results

Fake News Detector

Check whether a news article is real or fake

Aliens secretly controlling world governments

Predict

 Fake News (Exact Match)

Fake News Detector

Check whether a news article is real or fake

Experts warn that increasing conflicts across Ukraine, Middle East, and Asia could lead to a multi-dimensional global war situation

I

Predict

 Real News (Exact Match)

6. Advantages

The Fake News Detection System has several important advantages. It helps in automatically identifying whether a news article is real or fake, reducing the need for manual verification. This saves time and effort for users. The system provides quick and accurate results using machine learning techniques. It helps in reducing the spread of misinformation on social media and news platforms. The system is easy to use, as users only need to input the news text to get the prediction. It can also process large amounts of data efficiently, making it suitable for real-world applications. Additionally, it improves awareness among users by helping them verify information before believing or sharing it.

7. Limitations

Although the system is effective, it has certain limitations. The accuracy of the model depends heavily on the quality and size of the dataset used for training. If the dataset is limited or biased, the model may not perform well. The system may also struggle with new or unseen types of fake news that differ from training data. It mainly focuses on text-based analysis and may not consider images, videos, or other media formats. Continuous updates and retraining are required to maintain good performance over time.

8. Future Scope

The Fake News Detection System has a wide scope for future improvements and enhancements. In the current system, machine learning techniques are used to classify news as real or fake based on textual data. However, this system can be further improved to increase accuracy, efficiency, and real-world usability. In the future, more advanced techniques such as deep learning models (for example, LSTM and Transformer-based models like BERT) can be used to improve the accuracy of fake news detection. These models can better understand context and meaning in large text data compared to traditional machine learning algorithms. The system can also be expanded to work in real-time by integrating it with social media platforms and news websites. This will help in detecting fake news instantly as it is published or shared online. Another improvement can be the use of larger and more diverse datasets from different languages and regions, which will make the system more robust and globally applicable. Multilingual support can also be added so that news in different languages can be analyzed. Thus, the future scope of this project is very promising, and with further enhancements, it can become a powerful tool to reduce misinformation and improve information authenticity on digital platforms.

9. Conclusion

The Fake News Detection System successfully demonstrates how machine learning can be used to identify whether a news article is real or fake. In today's digital world, where information spreads rapidly through social media and online platforms, it has become difficult for users to verify the authenticity of news manually. This system helps solve that problem by automatically analyzing news content and providing accurate predictions.

The system uses a structured approach that includes data collection, preprocessing, feature extraction, model building, and training. By using machine learning algorithms, the system learns patterns from the dataset and effectively classifies news articles. The results show that the system is capable of detecting fake news with good accuracy and can be useful in reducing misinformation.

Overall, this project provides an efficient and reliable solution for fake news detection. It helps users verify information quickly and promotes awareness about false information on digital platforms. With further improvements and advanced techniques, the system can become even more accurate and useful in real-world applications.

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